Assignment — Week 05 · Day 25 (PM Session)

PG Diploma · AI-ML & Agentic AI Engineering · IIT Gandhinagar
Topics: Types of Machine Learning · Regression · Classification
Duration: 60–75 minutes
Submission: Submit on LMS with GitHub Repository Link

✅ **Part A — Concept Application (40%)**

🔹 **Problem Identification**

1️⃣ **A system predicts whether an email is spam or not using past labeled emails.**

Type: ✅ **Supervised Learning (Classification)**

Why?

- Uses labeled data (spam / not spam)
- Output is categorical

2️⃣ **A retail company groups customers based on purchasing behavior without predefined labels.**

Type: ✅ **Unsupervised Learning (Clustering)**

Why?

- No labeled output
- Model finds hidden patterns/groups

3️⃣ **A robot learns to walk by trying movements and receiving rewards.**

Type: ✅ **Reinforcement Learning**

Why?

- Learns via rewards and penalties
- Agent interacts with environment

4️⃣ **A model predicts house prices based on area, location, rooms.**

Type: ✅ **Supervised Learning (Regression)**

Why?

- Uses labeled data (price)
- Output is continuous (numeric)

5️⃣ **An e-commerce platform recommends similar products.**

Type: ✅ **Unsupervised Learning (Recommendation / Similarity-based)**

Why?

- Uses behavior patterns
- Often no explicit labels

🔹 Regression vs Classification

Criteria	Regression	Classification
Target Variable	Continuous	Categorical
Output Example	Price = ₹5,00,000	Spam / Not Spam
Algorithm Goal	Predict numeric value	Predict category

Justification:

- If target is numeric → Regression
- If target is category/label → Classification

🔹 Dataset Understanding (Supervised Example)

```
import pandas as pd

# Create small dataset
data = {
    'Area': [1000, 1500, 1800, 2400, 3000],
    'Rooms': [2, 3, 3, 4, 4],
    'Price': [50, 75, 90, 120, 150]  # in lakhs
}

df = pd.DataFrame(data)

# Separate features and target
X = df[['Area', 'Rooms']]
y = df['Price']

df
```

Explanation:

- **X (Features):** Area, Rooms
- **y (Target):** Price

✅ Part B — Stretch Problem (30%)

🔹 1️⃣ Regression Example

Problem:

Predict student marks based on hours studied.

```
data_reg = {
    'Hours_Studied': [2, 4, 6, 8, 10],
    'Marks': [40, 50, 65, 80, 95]
}

df_reg = pd.DataFrame(data_reg)
```

Explanation:

- **Input:** Hours_Studied
- **Output:** Marks (continuous)
- **Use Case:** Academic performance prediction

🔹 2️⃣ **Classification Example**

Problem:

Predict whether a customer will buy a product.

```
data_clf = {
    'Age': [22, 35, 26, 40, 28],
    'Salary': [20000, 50000, 30000, 80000, 35000],
    'Buy': [0, 1, 0, 1, 0]  # 0 = No, 1 = Yes
}

df_clf = pd.DataFrame(data_clf)
```

Explanation:

- **Input:** Age, Salary
- **Output:** Buy (Yes/No)
- **Use Case:** Marketing targeting

✅ **Part C — Interview Ready (20%)**

Q1 — What are types of Machine Learning?

There are three main types:

1️⃣ **Supervised Learning**

- Uses labeled data
- Examples: Regression, Classification

2️⃣ **Unsupervised Learning**

- No labeled output
- Example: Clustering

3️⃣ **Reinforcement Learning**

- Learns using rewards & penalties
- Example: Robotics, Game AI

Q2 — Coding Question

Separate features and target:

```
import pandas as pd

df = pd.read_csv("data.csv")  # Example dataset

X = df.drop('Target', axis=1)
y = df['Target']
```

Q3 — Difference Between Regression & Classification

Regression	Classification
Predicts numeric values	Predicts categories
Output is continuous	Output is discrete
Example: House price	Example: Spam detection

Interview-Level Answer:

Regression predicts **how much**, classification predicts **which class**.

✅ **Part D — AI-Augmented Task (10%)**

🔹 **Prompt Used:**
"Explain types of machine learning with real-world examples."

🔹 **AI Output (Documented)**
Machine learning is divided into:

1. **Supervised Learning**
   - Uses labeled data
   - Example: Email spam detection
2. **Unsupervised Learning**
   - No labels
   - Example: Customer segmentation
3. **Reinforcement Learning**
   - Learns via rewards
   - Example: Self-driving cars

🔹 **Evaluation of AI Output**

✅ Correctly defines three main types

✅ Gives real-world examples

⚠️ Could include more detail (algorithms, use cases)

⚠️ Does not explain regression vs classification clearly

Final Evaluation:
AI explanation is conceptually correct but requires deeper explanation for interview-level understanding.

GIT HUB REPOSITORY : https://github.com/Vicodwer/Day_25.git